HANSARD ONTOLOGY REPORT

Human Centred Systems

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Table of Contents

1. Introduction 3
2. Core Design Decisions 3
3. Ontology Structure 3
4. Reasoning 9
5. Ontograph 10
6. SPARQL Queries 11
   6.1 Query 1 - List all Hansard documents with their session, meeting, and sitting 11
   6.2 Query 2 - Show agenda items and contributions under them 12
   6.3 Query 3 - Retrieve contributions, speakers, and speech text 12
   6.4 Query 4 - Find every contribution that mentions a bill 13
   6.5 Query 5 – List all Members of Parliament 13
7. Linked Data Publication 14
   7.1 Repository structure 14
   7.2 Repository Contents 14
   7.3 Public URL and Reuse Value 14
8. Conclusion 15
Appendix A. OWL Fragment Summary 15

List of Tables

Table 1: Classes and Subclasses 3
Table 2: Object Properties 4
Table 3: Data Properties 6
Table 4: Named Individuals 7
Table 5: Key Restrictions Used in the Ontology 8
Table 6: Examples of Relationships Expected to be Inferred 9
Table 7: Structure of repository 14
Table 8: Contents of repository 14
Table 9: OWL fragment summary 15

Table of Figures

Figure 1: Classes and subclasses 4
Figure 2: Object properties 5
Figure 3: Data properties 7
Figure 4: Named individuals 8
Figure 5: AgendaItem example inferred in Protege 9
Figure 6: General ontograph of the ontology showing all classes and their relationships 10
Figure 7: Ontograph highlighting the central role of the *HansardDocument* class and its relationships to key entities 10
Figure 8: Detailed ontograph of the HansardDocument class illustrating its properties, constraints, and semantic structure 11
Figure 9: Query - List of Hansard documents with their session, meeting, and sitting 11
Figure 10: Query - Agenda items and contributions 12
Figure 11: Query - Contributions and speakers 12
Figure 12: Query- All contributions that mention a bill 13
Figure 13: Query - List of MPs 13

1. Introduction

This report documents the design and evaluation of a Hansard ontology created in Protégé. The ontology models a small part of parliamentary proceedings: sessions, meetings, sitting days, agenda items, debate contributions, members of parliament, bills, and the Hansard document that records the sitting. It focuses on creating the ontology, extending it with subclasses and constraints, checking it with a reasoner, querying it with SPARQL, and preparing it for publication as linked data.

The ontology contains 10 classes, 2 subclasses, 15 object properties, 21 data properties, and 10 named individuals. The data set is intentionally compact, but it is rich enough to demonstrate structure, constraints, inference, and capacity to be queried.

2. Core Design Decisions

The model is organised around the parliamentary workflow. A ParliamentSession contains meetings; a Meeting contains sitting days; a SittingDay contains agenda items; an AgendaItem contains debate contributions; and a HansardDocument describes the session, meeting, and sitting that it records. The ontology also separates documentary entities from real-world actors and parliamentary business. For example, HansardDocument is kept distinct from Person, AgendaItem, and LegislativeItem. The sample data centres on a single sitting day held on 12 March 2026, with one agenda item, one bill, two contributions, two MPs, one meeting, one session, and one Hansard document.

3. Ontology Structure

Table 1: Classes and Subclasses

Class	Superclass	Purpose
ParliamentSession	-	Formal parliamentary session
Meeting	-	Meeting within a parliamentary session
SittingDay	-	Dated sitting day linked to one meeting
AgendaItem	-	Scheduled unit of parliamentary business
DebateContribution	-	Speech or intervention in debate
Person	-	Human actor in the Hansard setting
MemberOfParliament	Person	Parliamentarian with constituency metadata
LegislativeItem	-	Formal business item introduced under an agenda item
Bill	LegislativeItem	Specialised legislative item
HansardDocument	-	Published documentary artefact

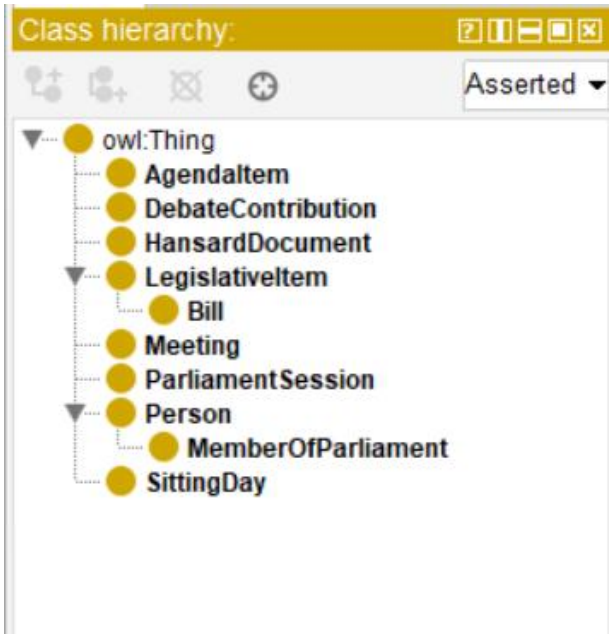

Figure 1: Classes and subclasses

Table 2: Object Properties

Property	Domain	Range	Purpose
describesSession	HansardDocument	ParliamentSession	Links the document to its session
describesMeeting	HansardDocument	Meeting	Links the document to its meeting
describesSitting	HansardDocument	SittingDay	Links the document to its sitting day
belongsToSession	Meeting	ParliamentSession	Meeting sits inside one session
hasMeeting	ParliamentSession	Meeting	Inverse of belongsToSession
belongsToMeeting	SittingDay	Meeting	Sitting sits inside one meeting
hasSittingDay	Meeting	SittingDay	Inverse of belongsToMeeting
belongsToSitting	AgendaItem	SittingDay	Agenda belongs to one sitting
containsAgendaItem	SittingDay	AgendaItem	Inverse of belongsToSitting
introducedInAgendaItem	LegislativeItem	AgendaItem	Anchors legislative item procedurally

occursUnderAgendaItem	DebateContribution	AgendaItem	Places a contribution under one agenda
hasContribution	AgendaItem	DebateContribution	Inverse of occursUnderAgendaItem
madeBy	DebateContribution	Person	Identifies the speaker
presentedBy	LegislativeItem	Person	Shows who presented the legislative item
mentionsBill	DebateContribution	Bill	Added realism property linking speech to bill mentioned

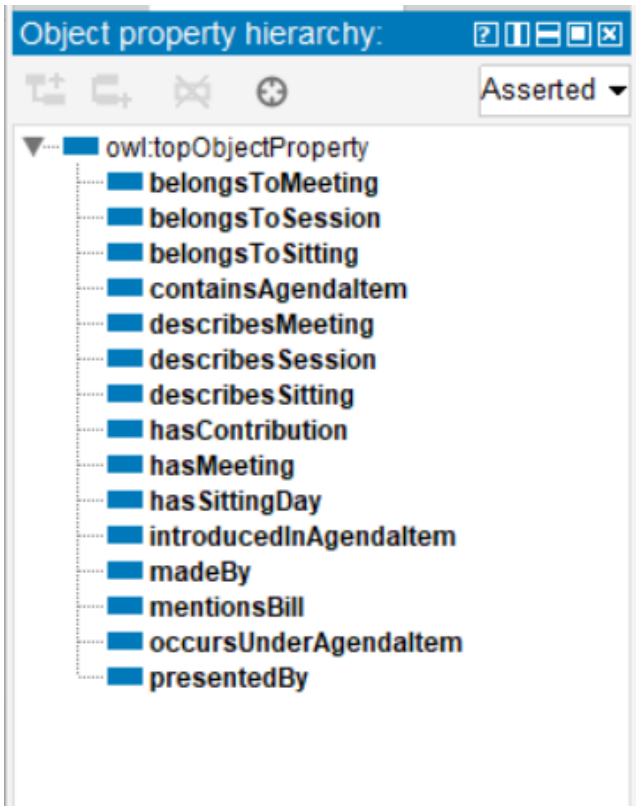

Figure 2: Object properties

Data properties provide literal values needed to identify and describe instances. They include constituencyName, sessionNumber, meetingNumber, sittingDate, agendaTitle, speechText, personName, itemTitle, hasISSN, and documentURL. These properties make the ontology suitable both for human-readable reporting and machine-readable querying.

Table 3: Data Properties

Data property	**Domain**	**Range**
hasTitle	HansardDocument	xsd:string
hasISSN	HansardDocument	xsd:string
hasEditionType	HansardDocument	xsd:string
publishedOn	HansardDocument	xsd:date
documentURL	HansardDocument	xsd:anyURI
sessionNumber	ParliamentSession	xsd:integer
sessionLabel	ParliamentSession	xsd:string
meetingNumber	Meeting	xsd:integer
meetingLabel	Meeting	xsd:string
sittingDate	SittingDay	xsd:date
sittingDayLabel	SittingDay	xsd:string
agendaTitle	AgendaItem	xsd:string
agendaOrder	AgendaItem	xsd:integer
itemTitle	LegislativeItem	xsd:string
itemNumber	LegislativeItem	xsd:string
readingStage	Bill	xsd:string
personName	Person	xsd:string
constituencyName	MemberOfParliament	xsd:string
mpCode	MemberOfParliament	xsd:string
speechOrder	DebateContribution	xsd:integer
speechText	DebateContribution	xsd:string

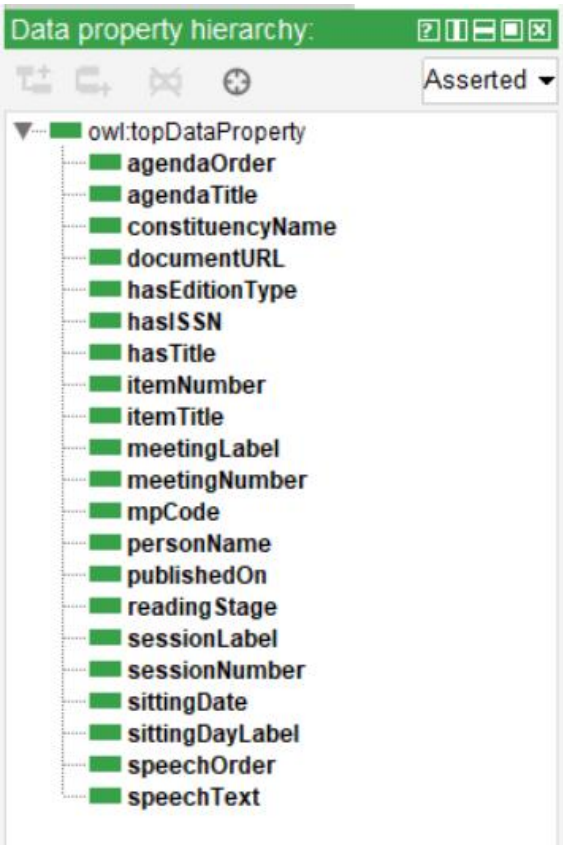

Figure 3: Data properties

This ontology contains 10 named individuals to demonstrate the complete chain from document metadata to speech content.

Table 4: Named Individuals

Individual	**Type**	**Role in example**
Hansard_2026_03_12	HansardDocument	Example Hansard issue
Session_51	ParliamentSession	Fifty-First Session
Meeting_51_02	Meeting	Second meeting in the session
Sitting_2026_03_12	SittingDay	Dated sitting
Agenda_NationalBudget	AgendaItem	Budget business slot
Bill_FiscalResponsibility2026	Bill	Specialised legislative item
Contribution_001	DebateContribution	First contribution
Contribution_002	DebateContribution	Second contribution
MP_Grace_Banda	MemberOfParliament	Presenter of the bill
MP_Samuel_Phiri	MemberOfParliament	Second contributor

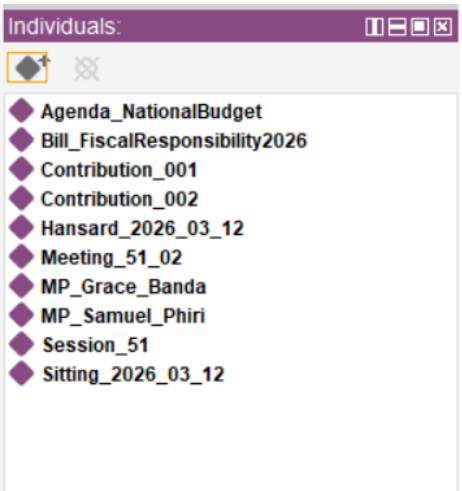

Figure 4: Named individuals

Table 5: Key Restrictions Used in the Ontology

Class	Restriction	Interpretation
AgendaItem	belongsToSitting exactly 1 SittingDay	Each agenda item is anchored to one sitting day
DebateContribution	madeBy exactly 1 Person	Each contribution has one speaker
DebateContribution	occursUnderAgendaItem exactly 1 AgendaItem	Each contribution belongs to one agenda item
HansardDocument	describesMeeting exactly 1 Meeting	Each document records one meeting
HansardDocument	describesSession exactly 1 ParliamentSession	Each document records one session
HansardDocument	describesSitting exactly 1 SittingDay	Each document records one sitting day
HansardDocument	hasISSN exactly 1 xsd:string	Each document contains one serial number
LegislativeItem	introducedInAgendaItem exactly 1 AgendaItem	Each legislative item is tied to one agenda slot
Meeting	belongsToSession exactly 1 ParliamentSession	Each meeting is part of one session
MemberOfParliament	mpCode min 1 xsd:string	Each MP must have a code
MemberOfParliament	constituencyName exactly 1 xsd:string	Each MP has one constituency
SittingDay	belongsToMeeting exactly 1 Meeting	Each sitting day belongs to one meeting
SittingDay	sittingDate exactly 1 xsd:date	Each sitting day has one date

4. Reasoning

A reasoner, HermiT, was used in Protégé to test consistency and to compute inferred knowledge. The ontology is expected to be consistent.

Table 6: Examples of Relationships Expected to be Inferred

Expected inferred statement	Why it should be inferred
Bill_FiscalResponsibility2026 rdf:type LegislativeItem	Bill is a subclass of LegislativeItem
MP_Grace_Banda rdf:type Person	MemberOfParliament is a subclass of Person
Session_51 hasMeeting Meeting_51_02	Inverse of belongsToSession
Meeting_51_02 hasSittingDay Sitting_2026_03_12	Inverse of belongsToMeeting
Sitting_2026_03_12 containsAgendaItem Agenda_NationalBudget	Inverse of belongsToSitting
Agenda_NationalBudget hasContribution Contribution_001 / Contribution_002	Inverse of occursUnderAgendaItem

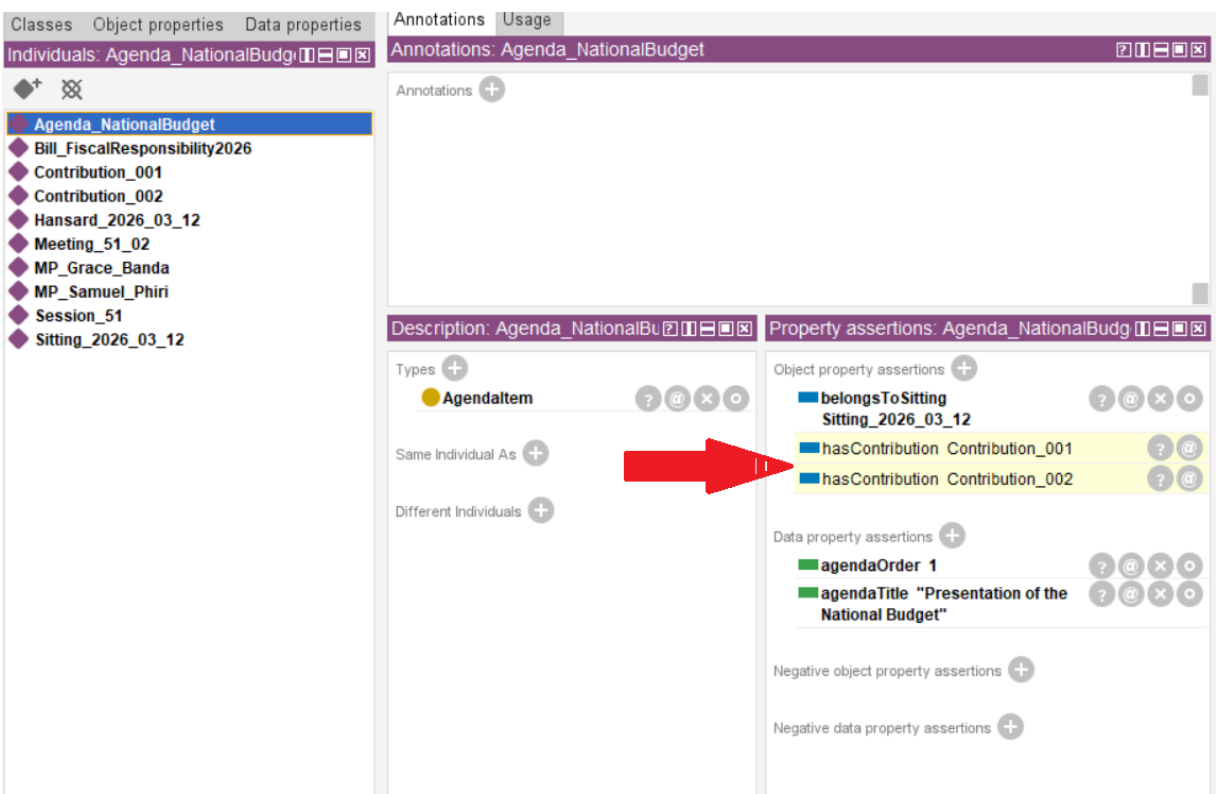

Figure 5: AgendaItem example inferred in Protege

5. Ontograph

In this report, three complementary ontographs are used: first, a general ontograph that presents the complete ontology, showing all defined classes (e.g., *Person*, *DebateContribution*, *AgendaItem*, *SittingDay*) and their interconnections; second, a focused ontograph that highlights the central role of the *HansardDocument* class, illustrating how it links to key entities such as sittings, meetings, and sessions; and third, a detailed class-level ontograph that exposes the internal characteristics of *HansardDocument*, including its properties, superclass relationships, and constraints (e.g., cardinality restrictions and disjointness), thereby clarifying its semantic definition within the ontology.

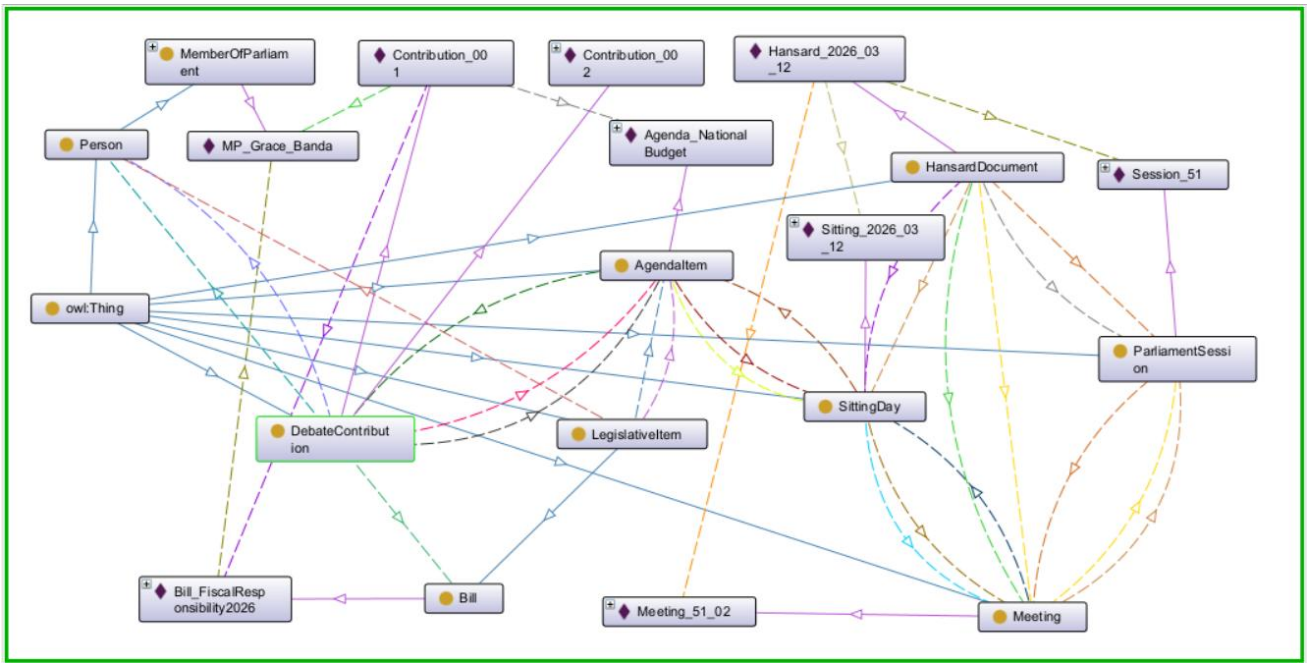

Figure 6: General ontograph of the ontology showing all classes and their relationships

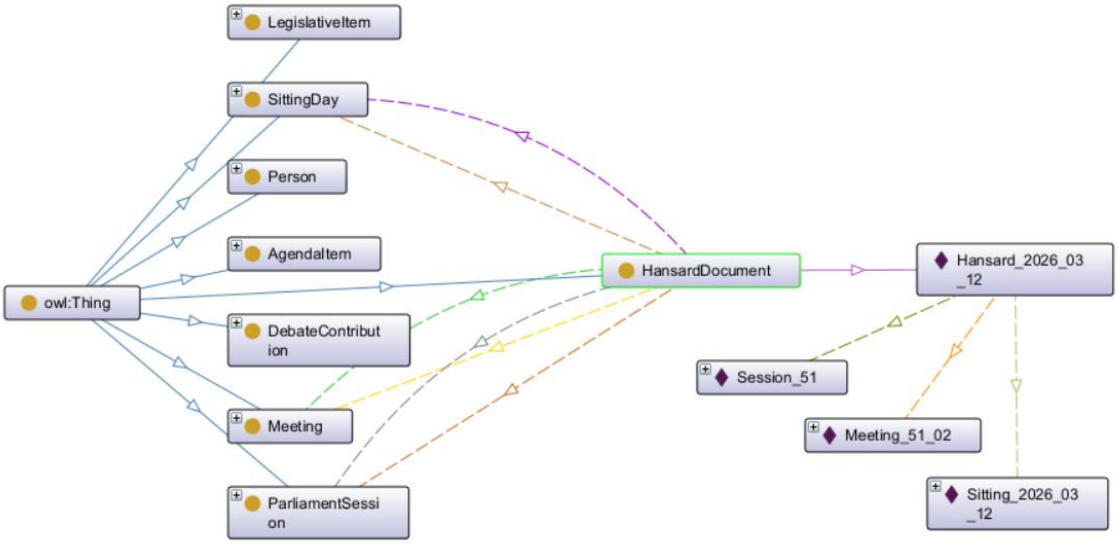

Figure 7: Ontograph highlighting the central role of the *HansardDocument* class and its relationships to key entities

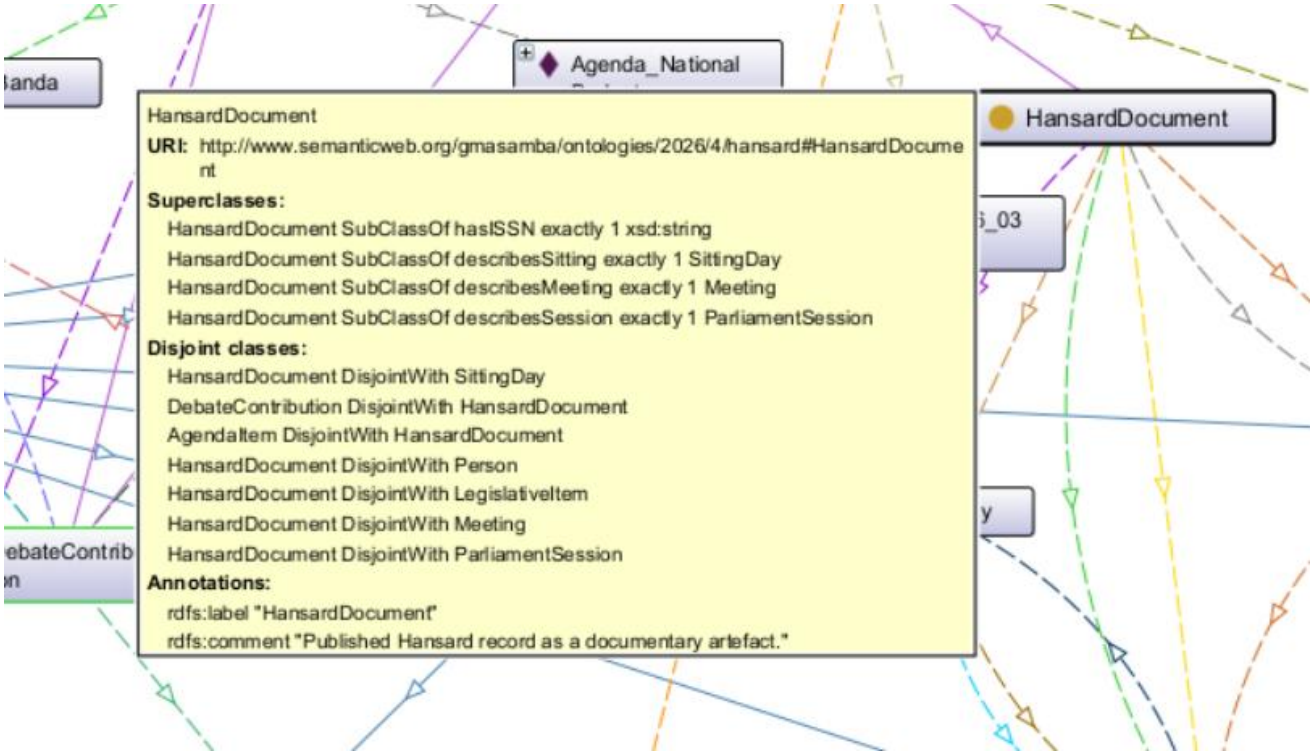

Figure 8: Detailed ontograph of the HansardDocument class illustrating its properties, constraints, and semantic structure

6. SPARQL Queries

6.1 Query 1 - List all Hansard documents with their session, meeting, and sitting

SPARQL query:

```
PREFIX : <http://www.semanticweb.org/gmasamba/ontologies/2026/4/hansard#>
SELECT ?doc ?session ?meeting ?sitting
WHERE {
 ?doc a :HansardDocument ;
    :describesSession ?session ;
    :describesMeeting ?meeting ;
    :describesSitting ?sitting .
}
```

doc	session	meeting	sitting
Hansard_2026_03_12	Session_51	Meeting_51_02	Sitting_2026_03_12

Figure 9: Query - List of Hansard documents with their session, meeting, and sitting

6.2 Query 2 - Show agenda items and contributions under them

SPARQL query:

```
PREFIX : <http://www.semanticweb.org/gmasamba/ontologies/2026/4/hansard#>
SELECT ?agenda ?contribution
WHERE {
  ?contribution :occursUnderAgendaItem ?agenda .
}
ORDER BY ?agenda ?contribution
```

agenda	contribution
Agenda_NationalBudget	Contribution_001
Agenda_NationalBudget	Contribution_002

Figure 10: Query - Agenda items and contributions

6.3 Query 3 - Retrieve contributions, speakers, and speech text

SPARQL query:

```
PREFIX : <http://www.semanticweb.org/gmasamba/ontologies/2026/4/hansard#>
SELECT ?contribution ?speaker ?text
WHERE {
  ?contribution a :DebateContribution ;
          :madeBy ?speaker ;
          :speechText ?text .
}
ORDER BY ?contribution
```

contribution	speaker	text
Contribution_001	MP_Grace_Banda	"I beg to present the Fiscal Responsibility Bill, 2026 for First Reading."
Contribution_002	MP_Samuel_Phiri	"The House should note the accountability safeguards proposed in the bill."

Figure 11: Query - Contributions and speakers

6.4 Query 4 - Find every contribution that mentions a bill

SPARQL query:

```
PREFIX : <http://www.semanticweb.org/gmasamba/ontologies/2026/4/hansard#>
SELECT ?contribution ?bill
WHERE {
  ?contribution :mentionsBill ?bill .
}
```

contribution	bill
Contribution_001	Bill_FiscalResponsibility2026
Contribution_002	Bill_FiscalResponsibility2026

Figure 12: Query- All contributions that mention a bill

6.5 Query 5 – List all Members of Parliament

SPARQL query:

```
PREFIX h: <http://www.semanticweb.org/gmasamba/ontologies/2026/4/hansard#>

SELECT ?mp ?name ?code ?constituency
WHERE {
  ?mp rdf:type h:MemberOfParliament ;
     h:personName ?name ;
     h:mpCode ?code ;
     h:constituencyName ?constituency .
}
ORDER BY ?name
```

mp	name	code	constituency
MP_Grace_Banda	"Hon. Grace Banda"	"MP-051-014"	"Lilongwe Central"
MP_Samuel_Phiri	"Hon. Samuel Phiri"	"MP-051-027"	"Mzuzu South"

Figure 13: Query - List of MPs

7. Linked Data Publication

The ontology is uploaded to GitHub and GitHub Pages, with a public URL and a brief explanation of reuse. The ontology includes a documentURL value in the HansardDocument individual: https://20115492.github.io/hansard-ontology/. A GitHub repository can be found at: https://github.com/20115492/hansard-ontology.

7.1 Repository structure

```
hansard-ontology/
  ├── Hansard.owl
  ├── Hansard_Report.pdf
  ├── README.md
  └── index.html
```

Table 7: Structure of repository

7.2 Repository Contents

File	Role in publication
Hansard.owl	The ontology itself, published as the core semantic asset
Hansard_Report.pdf	PDF version of the assignment report
README.md	Repository overview, ontology purpose, usage notes, and publication link
index.html	Simple landing page for GitHub Pages publication and download access

Table 8: Contents of repository

7.3 Public URL and Reuse Value

A public GitHub Pages URL makes the ontology discoverable on the web. The HansardDocument instance points directly to a live published resource at https://20115492.github.io/hansard-ontology/. This makes the ontology to be addressable, shareable, and reusable.

The ontology could be reused in the following ways:

- as a teaching example for ontology engineering and Protégé-based modelling;
- as a starter model for parliamentary information systems, debate indexing, or legislative archives;
- as a basis for extending into richer linked data on committees, motions, voting, ministers, or political parties;
- as a semantic layer for search and analytics over Hansard documents and parliamentary speech data.

8. Conclusion

The Hansard ontology contains classes, subclasses, individuals, and object properties; it uses explicit disjointness axioms and multiple qualified cardinality restrictions; it is suitable for reasoning; and it supports meaningful SPARQL queries. It models a parliamentary scenario with a document-to-event structure. The use of inverse properties, subclass relations, publication metadata, and speech-to-bill links makes the ontology both coherent and extensible.

Appendix A. OWL Fragment Summary

Item	Value
Ontology IRI	http://www.semanticweb.org/gmasamba/ontologies/2026/4/hansard
Serialization	RDF/XML
Class count	10
Named individual count	10
Main inverse property pairs	belongsToSession/hasMeeting; belongsToMeeting/hasSittingDay; belongsToSitting/containsAgendaItem; occursUnderAgendaItem/hasContribution
Key restrictions	Exactly one session per meeting; exactly one meeting per sitting; exactly one sitting per agenda item; exactly one speaker and one agenda item per contribution

Table 9: OWL fragment summary